

Niranjan Saravanakumar

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Education

K S Rangasamy College of Technology	2023 – 2027
Bachelor of Technology in Information Technology	CGPA: 8.63 / 10.0
Shri Janani Matric Higher Secondary School	2022 – 2023
Higher Secondary Education	Percentage: 89%

Experience

ABB Global Industries and Services Private Limited	August 2025 – May 2026
Software Engineering Intern	Bangalore, India
- Increased automated test coverage by 80% by authoring 300+ unit and integration tests and building Playwright UI automation using Python, Codegen, Chromium, PyTest, Angular, Jasmine, Karma, and C#. - Cut release validation time by 45% by redesigning testing workflows around Docker containers and Azure DevOps CI/CD pipelines. - Reduced critical code issues by 70% by triaging and resolving 50+ SonarQube vulnerabilities across UI and API codebases. - Shortened defect identification time by 35% by building automation tooling for .NET-based systems in Python and C#, enabling faster root-cause analysis through automated test execution. - Partnered with an 8-person cross-functional team of developers and QA engineers to align testing standards and streamline deployment workflows, reducing review cycle friction.	

Projects

CodeCompass FastAPI, Python, Groq LLaMA 3.3, Axios, React
- Built a FastAPI backend that fetches repository files via the GitHub API and uses Groq LLaMA 3.3 to auto-generate structured onboarding docs. - Implemented stateful multi-turn chat with Pydantic-validated routes and 24-hour caching, turning any public GitHub repo into a queryable knowledge base. - Developed a React 19 + Vite 8 frontend with Markdown rendering for a smooth chat and doc-reading experience.
SentiAI Python, Flask, scikit-learn, NLTK
- Built and deployed an end-to-end sentiment analysis web application using Flask, training a Logistic Regression model with TF-IDF features on the Sentiment140 dataset (1.6M tweets). - Designed a REST API (POST /api/analyze) for real-time sentiment prediction, returning JSON responses with confidence scores generated via <code>predict_proba()</code>. - Engineered an NLP preprocessing pipeline using NLTK (text normalization, stopwords removal, Porter stemming), serialized model artifacts with pickle, and implemented automated testing with pytest.

Technical Skills

Programming Languages: Java, Python
Frameworks & Libraries: React.js, Flask, Jasmine, Karma
DevOps & Cloud: Azure DevOps, GitHub Actions, CI/CD Pipelines
Tools & Platforms: Git, VS Code, Visual Studio, SonarQube
Databases: SQL

Certifications

Oracle Certified Foundations Associate – Oracle University
Design & Implementation of Human-Computer Interfaces – NPTEL Online

Achievements

- Participated in the ABB Hackathon 2025, collaborating in a cross-functional team to design and prototype an innovative software solution.
- Solved 300+ LeetCode problems, strengthening problem-solving skills in Data Structures and Algorithms using Java.